FIITJEE Students From All Programs have bagged 34 in Top 100, 66 in Top 200 and 174 in Top 500 All India Ranks. FIITJEE Performance in JEE (Advanced), 2014:
2521 FIITJEE Students from Classroom / Integrated School Programs & 3579 FIITJEE Students from All Programs have qualified in JEE (Advanced), 2014.

FIITJEE

# JEE(Advanced)-2015

ALL INDIA TEST SERIES

## ANSWERS, HINTS & SOLUTIONS
## CRT– I
## (Paper-2)

| Q. No. | PHYSICS | CHEMISTRY | MATHEMATICS |
|---|---|---|---|
| 1. | **D** | **A** | **C** |
| 2. | **B** | **A** | **C** |
| 3. | **B** | **C** | **C** |
| 4. | **B** | **C** | **B** |
| 5. | **D** | **A** | **D** |
| 6. | **C** | **D** | **A** |
| 7. | **C** | **D** | **C** |
| 8. | **D** | **A** | **B** |
| 9. | **D** | **A** | **A** |
| 10. | **C** | **C** | **C** |
| 11. | **D** | **C** | **A** |
| 12. | **B** | **B** | **C** |
| 13. | **B** | **B** | **C** |
| 14. | **C** | **C** | **A** |
| 15. | **D** | **B** | **C** |
| 16. | **C** | **A** | **B** |
| 17. | **A** | **A** | **C** |
| 18. | **C** | **A** | **B** |
| 19. | **D** | **A** | **D** |
| 20. | **B** | **B** | **D** |

# Physics

## PART – I

### SECTION – A

2. $6 = V\cos\theta\, t$ ...(1)

$3 = V\sin\theta\, t - \frac{1}{2}gt^2$ ...(2)

$24 = \frac{V^2\sin 2\theta}{g}$ ...(3)

Solving the equation we will get the result

3. $\frac{1}{2}kx^2 = \frac{1}{2}mv^2$

$x = \sqrt{\frac{mv^2}{k}}$

4. $F = \mu mg$ use concept of toppling

5. Use concept of conservation of energy $E_i = E_f$

6. $a = \frac{mg - F_v}{m}$

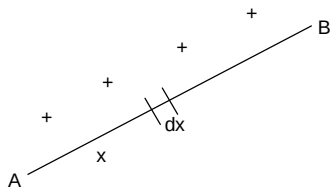
$v = u + at$

7. Use $B = \frac{\mu_0}{4\pi}\cdot\frac{2i}{r}$

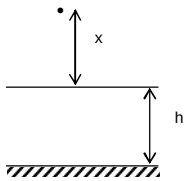
8. $\vec{M} = iA$

$A = \pi x^2, i = \frac{\lambda dx}{t}, t = \frac{2\pi}{\omega}$

Place the value in M = iA




9. $x + \frac{h}{\mu}$ = distance of final image of bird from bird itself



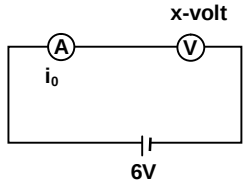

11. $\vec{r} = \hat{i}(t^2 - 4t) + \hat{j}(-t^3)$

Differentiation twice to get $\vec{a} = \hat{i}(2) - 6t\,\hat{j}$

$\vec{\tau} = \vec{r} \times \vec{F}$

12. $\omega = \int F \cdot dr$

15. $(6 - x)\, i_0 R_A$ and $\left(6 - \frac{x}{3}\right) = (3i_0)R_A$




$\Rightarrow \quad \frac{6-x}{6-\frac{x}{3}} = \frac{1}{3}$

$\Rightarrow \quad x = \frac{9}{2}V$

$\Rightarrow$ after connection = $\frac{3}{2}V$

16. $i_0 \times R_V = x$ and $(3i_0)\left(\frac{3R_v}{3+R_v}\right) = \frac{x}{3}$

$\Rightarrow \quad i_0 = \frac{3}{16}A$

17. (P) $V_{cm} = (R - r)\omega$

$\therefore$ rotational $E = \frac{1}{4}m(R-r)^2\omega^2$

(Q) total kinetic energy = $\frac{1}{2}mv_{cm}^2 + \frac{1}{2}I_{cm}\omega^2 = \frac{3}{4}m(R-r)^2\omega^2$

(R) $\vec{a}_{cm} = (R-r)\alpha\hat{i} + (R-r)\omega^2\hat{j}$

18. $\vec{L}_{\text{any point}} = \vec{L}_{\text{cm, that point}} + \vec{L}_{\text{about CM}}$

(P) $\rightarrow \frac{3}{2}mvr\left(-\hat{k}\right)$

(Q) $\rightarrow$ (1)

$K = \frac{1}{2}mv_{cm}^2 + \frac{1}{2}I_{cm}\omega^2$

$\Rightarrow$ (R) $\rightarrow \frac{3}{4}mv^2$

19. Slipping between 2 and 8

(P) $a_{2\,kg} = \frac{14-12}{2} = 1$

(Q) $a_{8\,kg} = \frac{12-10}{8} = \frac{1}{4}$

(R) $0.6 \times 2 \times 10 = 12$

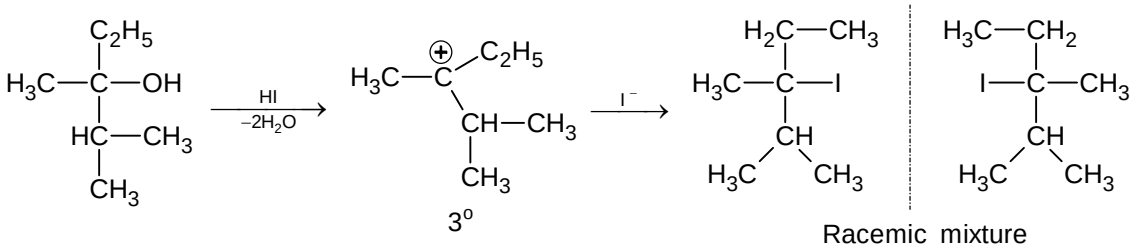
20. Basic combination of lenses.

# Chemistry

## PART – II

## SECTION – A

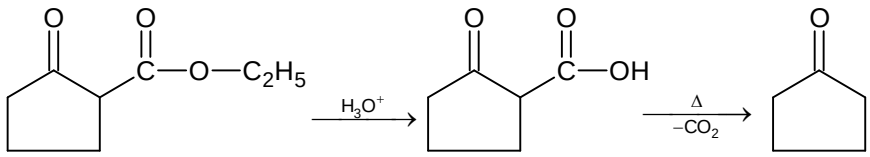
1.




3°

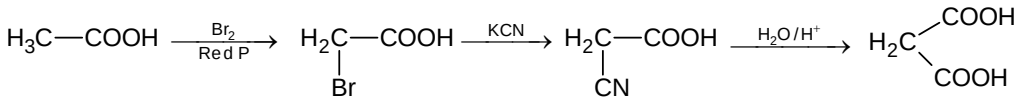
Racemic mixture

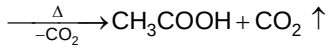
2.



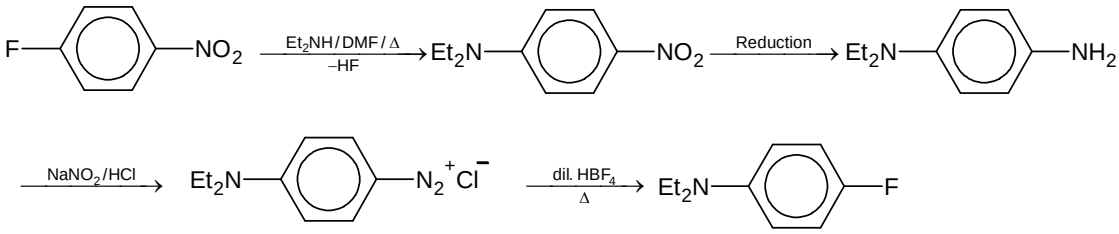

β – ketoacid

3.

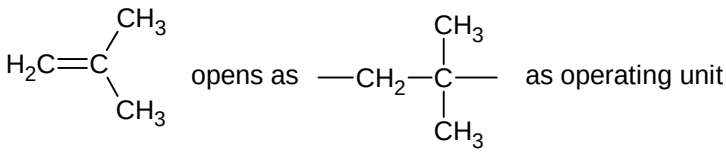


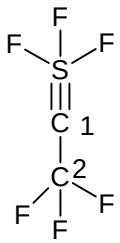

4.




5.




6.




7. $XF_4$-type molecule have zero dipole moment when it is either tetrahedral or square planar.

8. $\frac{r_{A^+}}{r_{B^-}} = 0.225 \Rightarrow \frac{22.5}{r_{B^-}} = 0.225$

$\Rightarrow r_{B^-} = 100$ pm

9. $\Delta G^o = 2\Delta G^o_{ZnO} + 2\Delta G^o_{SO_2} - 2\Delta G^o_{ZnS}$

= (–616–408)–(–293) = –731 J

10. Expansion of real gas above its inversion temperature shows heating effect.

11. $sp^3d^2 \rightarrow$ it involves $d_{x^2-y^2}$ and $d_{z^2}$ orbitals in hybridisation.

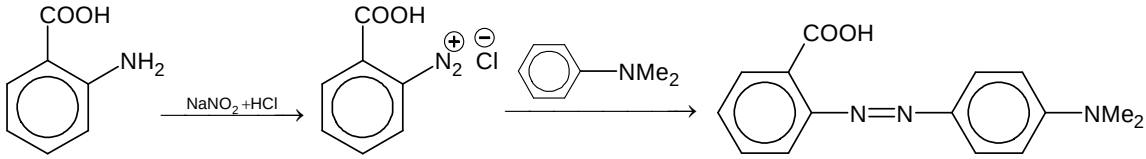
12. $sp^3d^3 \rightarrow$ It involves $d_{xy}$, $d_{yz}$, $d_{zx}$ d-orbtials in hybridisation.

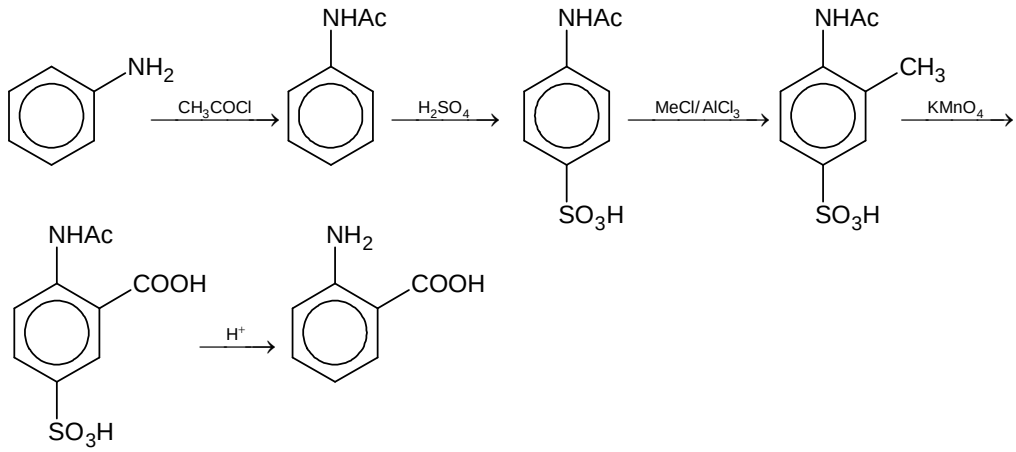
13. $\Delta G^o = -nFE^o$

$= -2 \times 96500 \times 1.10 = -212.3$ KJ

14. $\left(\frac{dE}{dT}\right)_P = \frac{\Delta S}{nF}$

$\Rightarrow 1.5 \times 10^{-4} = \frac{\Delta S}{2 \times 96500} \Rightarrow \Delta S = 28.95\ JK^{-1}\ mol^{-1}$

15. 


16. 


17. Fact.

19. Read the text.

20. –COOH, $–NO_2$ groups are deactivating groups and $–CH_3$, $–NMe_2$ are activating group. Friedel Craft's acylation not take place with the deactivated benzene compound $\bar{N}H_2$ is nucleophile.

# Mathematics

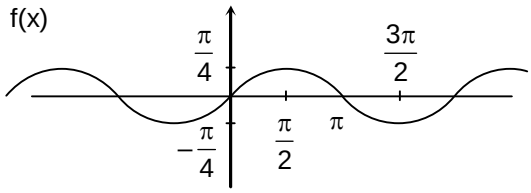
## PART – III

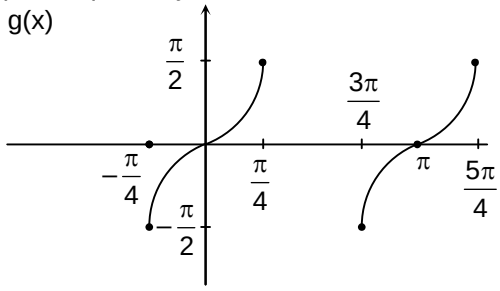
## SECTION – A

1. At $x = 0, \pm\pi, \pm 2\pi$
LHS = RHS
Now let $f(x) = \tan^{-1}(\sin x)$ and $g(x) = \sin^{-1}(\tan x)$

$$f'(x) = \frac{\cos x}{1+\sin^2 x} \text{ and } g'(x) = \frac{\sec^2 x}{\sqrt{1-\tan^2 x}} \geq \frac{\cos x}{\sqrt{1-\tan^2 x}} \geq \frac{\cos x}{1+\sin^2 x} = f'(x)$$

Hence, $f'(x) \leq g'(x)\ \forall\ x \in$ domain. So f(x) and g(x) are equal only at $x = n\pi$




2. $P(x) - 2008 + x = x(x-1)(x-2)(x-3) \ldots\ldots (x-2008)$
Put $x = 2009$

3. $\therefore f'(x) = 3x^2 + 2x + 100 + 7\cos x$
$= 3x^2 + 2x + 93 + 7(1 + \cos x)$

$$= 3\left\{\left(x+\frac{1}{3}\right)^2\right\} + \frac{278}{3} + 7(1+\cos x) > 0$$

$\therefore$ f(x) is increasing function $\Rightarrow f(1) < f(2) < f(3)$
(Let $a < b < c$)
Now $g(y) = (y-b)(y-c) + 2(y-c)(y-a) + 3(y-a)(y-b) = 0$
$\therefore g(a) > 0, g(b) < 0$ and $g(c) > 0$

4. $\because \quad \frac{C_0}{x} - \frac{C_1}{x+1} + \frac{C_2}{x+2} - \frac{C_3}{x+3} + \ldots\ldots + (-1)^n \frac{C_n}{x+n} = \frac{n!}{x(x+1)(x+2)\ldots\ldots(x+n)}$

$$\therefore \text{ At } x = 2,\ u_n = \frac{C_0}{2} - \frac{C_1}{3} + \frac{C_2}{4} - \frac{C_3}{5} + \ldots\ldots = \frac{1}{(n+1)(n+2)} = \frac{1}{n+1} - \frac{1}{n+2}$$

Hence, $s_n = u_1 + u_2 + u_3 + \ldots\ldots + u_n = \left[\frac{1}{2} - \frac{1}{n+2}\right]$

5. (1) ad = 1, bc = 4 (2) ad = 1, bc = 2
(3) ad = 2, bc = 1 (4) ad = 2, bc = 4
(5) ad = 4, bc = 1 (6) ad = 4, bc = 2

6. Centroid = circumcentre = $(x_1, y_1) = \left[\frac{\sum a\cos\alpha}{3}, \frac{\sum b\sin\alpha}{3}\right]$

$$\left(\frac{9x_1^2}{a^2} + \frac{9y_1^2}{b^2} - 3\right) = 2\left(\sum\cos\alpha\cos\beta + \sum\sin\alpha\sin\beta\right)$$

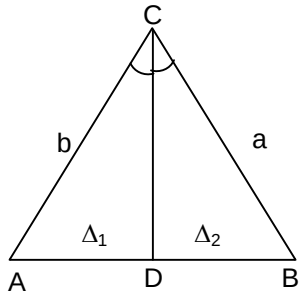
7. $\Delta = \Delta_1 + \Delta_2$

$\Delta = \frac{1}{2} ab \sin c = ab \sin\frac{C}{2}\cos\frac{C}{2}$

$\Delta_1 + \Delta_2 = \frac{1}{2} 6b \sin\frac{C}{2} + \frac{1}{2} 6a \sin\frac{C}{2}$

$\Delta = \Delta_1 + \Delta_2$

$\frac{1}{a} + \frac{1}{b} = \frac{1}{9} \Rightarrow \frac{ab}{a+b} = 9$




8. $M(x) = \int \frac{e^x\left(e^{2x} - 2\right)}{e^{4x} + 8e^{2x} + 4} dx$

Put $e^x = t$

9. $f(x) = x^2 + \int_0^x e^{-t} f(x-t)dt = x^2 + \int_0^x e^{-(x-t)} f(t)dt = x^2 + e^{-x}\int_0^x e^t f(t)dt$

$\therefore e^x f(x) = x^2 e^x + \int_0^x e^t f(t)dt$

Differentiating both sides w.r.t. x

$e^x f(x) + e^x f^1(x) = 2xe^x + x^2 e^x + e^x f(x)$

$\therefore f^1(x) = 2x + x^2$

$\therefore f(x) = x^2 + \frac{x^3}{3} + c$

10. Given differential equation can be written as

$\frac{xdy + ydx}{(xy-1)^2} = \frac{dx}{x^2}$

Integrating both sides

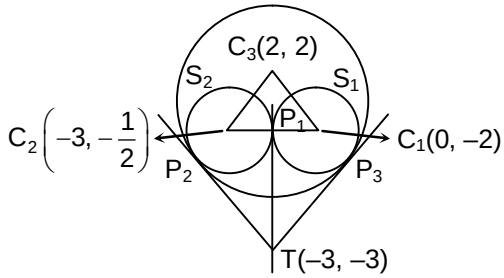
$-\frac{1}{(xy-1)} = -\frac{1}{x} + c \Rightarrow \frac{1}{xy-1} = \frac{1}{x} + c$

11.-12. Shown in figure

$P_1(-2, -1)$

$P_2(-4, -1)$

$P_3(-1, -4)$




13. $f''(c_1) > f''(c_2)$

$f''(c_1) > 0$ and $f''(c_2) < 0$

$\Rightarrow$ at $x = c_1$, f has local minimum and at $x = c_2$, f has local maximum

14. See above solution

15. $\phi(8n + 4) = \phi(4(2n + 1)) = \phi(4)\ \phi(2n + 1) = \phi(2^2)\ \phi(2n + 1) = 2\phi(2n + 1)$

16. $\phi(p^n)$ is odd $\Rightarrow p^{n-1}(p - 1)$ is odd

$\therefore$ p is prime, the only value p can take is p = 2
$\therefore$ $\phi(2^n)$ is odd

17. (P) $f'(x) = 2x + g'(1)$, $f''(x) = 2$
$g'(x) = 2x + f'(2)$, $g''(x) = 2$
at $x = 1$, $f'(1) = 2 + g'(1)$ and $g'(1) = 2 + f'(2)$
$\Rightarrow f'(1) = 4 + f'(2)$
(Q) $\therefore 2f(x) f(y) = f(x - y) + f(x + y)$ ..... (1)
Replacing x by y and y by x, then
$2f(y) f(x) = f(y - x) + f(y + x)$ ..... (2)
From equation (1) and (2), we get $f(x - y) = f(y - x)$
Put y = 0, then differentiate
(R) $f'(x) = 3x^2 + 2xf'(1) + f''(2)$
$\Rightarrow f'(1) = 3 + 2f'(1) + f''(2)$
$\Rightarrow f'(1) + f''(2) = -3$
$\Rightarrow$ and $f''(x) = 6x + 2f'(1)$
$\therefore f''(2) = 12 + 2f'(1)$
$\therefore -2f'(1) + f''(2) = 12$
(S) $n(a + b)^{n-1} = na^{n-1} + nb^{n-1}$

18. (P) Put t + 2 = z
(Q) Put 3t = z
(R) Integrate by part
(S) Put $\frac{1}{t} = z$

19. (P) $P\left(\frac{A_4}{B}\right) = \frac{\frac{1}{4}\times 1}{\frac{1}{4}\times\frac{1}{10}+\frac{1}{4}\times\frac{3}{10}+\frac{1}{4}\times\frac{6}{10}+\frac{1}{4}\times 1} = \frac{1}{2}$

(Q) Required probability = $\frac{3\times {}^5C_3 + 5^3}{{}^{15}C_3} = \frac{31}{91}$

(R) Required probability = $\frac{3!(2)}{4\times 4\times 4} = \frac{3}{16}$

(S) $P(A) = \sum_{i=1}^{3} P(E_i)\cdot P\left(\frac{A}{E_i}\right) = \frac{1}{3}\times 1 + \frac{1}{3}\times\frac{1}{2} + \frac{1}{3}\times\frac{1}{6} = \frac{5}{9}$

Thus, $P\left(\frac{E_1}{A}\right) = \frac{\frac{1}{3}\times 1}{\frac{5}{9}} = \frac{3}{5}$, $P\left(\frac{E_2}{A}\right) = \frac{\frac{1}{3}\times\frac{1}{2}}{\frac{5}{9}} = \frac{3}{10}$ and $P\left(\frac{E_3}{A}\right) = \frac{\frac{1}{3}\times\frac{1}{6}}{\frac{5}{9}} = \frac{1}{10}$

20. (P) $\sum_{r=1}^{10}(r-\omega)\left(r-\omega^2\right) = \sum_{r=1}^{10}\left\{r^2 - r(-1) + 1\right\}$

(Q) $\alpha^4 + \beta^4 + \alpha^{-1}\beta^{-1} = \omega^4 + \omega^8 + \omega^{-1}\omega^{-2} = \omega + \omega^2 + 1 = 0$

(R) $N^r = 1 + \cos\theta + i\sin\theta = 2\cos\frac{\theta}{2}\left[\cos\frac{\theta}{2} + i\sin\frac{\theta}{2}\right] = 2\cos\frac{\theta}{2}\cdot e^{i\theta/2}$

$D^r = -i^2\sin\theta + i(1 + \cos\theta) = i$ [conjugate of $N^r$] $= i2\cos\frac{\theta}{2}\cdot e^{-i\theta/2}$

(S) $\frac{\pi}{3} + \frac{\pi}{3^2} + \frac{\pi}{3^3} + ..... \infty = \frac{\pi}{2}$